15.05.2025 (E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: January 2025–April 2025 Examination: ESE Examination	Duration:3 Hrs.
Programme code: h54 Programme: Honours_Data Science & Analytics	Class: SY	Semester:IV(SVU 2020)
Name of the Constituent College: K. J. Somaiya School of Engineering	Name of the department: Electronics and Computer Engineering	
Course Code: 116h54C401	Name of the Course: Applied Data Science	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Explain the impact of applying data science in business scenarios.	5
ii)	Calculate the range, mode, median, mean of the given data. 5,10,3,6,4,8,9,3,15,2,9,4,19,11,4	5
iii)	What are the applications of supervised machine learning algorithms in modern business?	5
iv)	Compare classification with regression with an example.	5
v)	Describe cost function. Explain how to minimize cost function.	5
vi)	Explain radial basis function.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain types of Exploratory Data Analysis	5
ii)	Explain cosine and euclidean distance measures with example.	5
	OR	
Q2 A	Why is data preprocessing important? Describe data integration method in detail.	10
Q 2 B	Solve any **One**	10
i)	Write a short note on Jaccard distance with applications . Calculate the Jaccard distance for the following. A={3,4,5,6} B={5,6,7,8}	10
ii)	Explain Data discretization. Divide the given data into equal width intervals using Equal-Width Binning. Input=[5,10,50,72,92,104,215], Bins:3	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain precision, recall, F1 score and accuracy. Calculate precision, recall, F1 score and accuracy for the following cases in the data set. True Positive=30, True Negative=20, False Positive=10, False Negative=15.	10
ii)	Explain bayes theorem and bayes classifier with an example.	10
iii)	What do you mean by Decision tree? How does the Decision tree algorithm work? Explain the attribute selection measure information gain.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Explain k-means clustering with example.List advantages and disadvantages of k-means clustering.	10
ii)	Describe expectation maximization algorithm. How does the expectation maximization algorithm work?	10
iii)	Explain text processing and text summarization.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	What are Descriptive Statistics? Explain central tendency, variability and distribution.	5
ii)	Distinguish between overfitting and underfitting. How can it affect model generalization?	5
iii)	Define 1. Variance 2. Standard Deviation. Consider the following dataset and calculate the variance and standard deviation. {10,15,12,18,20}	5
iv)	Describe the significance of Kernel functions in SVM. List any two kernel functions.	5
v)	Explain rule- based classification with example.	5
vi)	List any five applications of machine learning clustering algorithms	5